

Deploying KIVA on Your Machine

A Quick Walkthrough

MA AI Hub x SundAI Pre-Session

May 6, 2026

KIVA is built on **Riverst**

Original Project

Riverst by Sensein Lab

<https://github.com/sensein/riverst>

Platform for building interactive user-avatar conversations with speech analysis

This Fork

KIVA (Knowledge Integration and Vocabulary Acquisition)

<https://github.com/ebaenamar/KIVA>

Adds easy installation scripts and deployment guides for educational activities

What You Need

Hardware & OS

- macOS (Apple Silicon) or Ubuntu Linux
- **Windows:** Use GitHub Codespaces

Software

- Python 3.11+ (or conda)
- Node.js 16+
- Git
- Chrome 134+ (for WebRTC)

API Keys

- **Required:** OpenAI API key
- **Optional:** Hugging Face token

Get your OpenAI key

<https://platform.openai.com/api-keys>

Windows Users

See slide on GitHub Codespaces

Step 1: Clone the Repo

```
git clone https://github.com/ebaenamar/KIVA.git
```

```
cd KIVA
```

What you get

src/server/ – Python backend | src/client/react/ – React frontend

Step 2: Server Setup

Create a conda environment and install dependencies:

```
cd src/server
```

```
conda create -n riverst python=3.11
```

✓ one-time only

```
conda activate riverst
```

↻ every time

```
conda install -c conda-forge ffmpeg
```

✓ one-time only

```
pip install -r requirements.txt
```

✓ one-time only

No conda?

Use venv instead: `python -m venv venv && source venv/bin/activate` Install ffmpeg via your system package manager, then `pip install -r requirements.txt`

Step 3: Configure Environment

Server (src/server/.env):

- 1 `cp env.example .env`
- 2 Open `.env` and set:
 - `OPENAI_API_KEY="sk-your-key-here"`  **required**
 - `HF_TOKEN="hf_..."` optional
 - Leave everything else as-is

Client (src/client/react/.env):

- 1 `cp env.example .env`
- 2 Change `VITE_API_PROXY_TARGET` to `http://localhost:7860`
- 3 Leave everything else as-is

Step 4: Client Setup

```
cd src/client/react
```

```
npm install
```

One command, that's it.

Node handles everything via `package.json`.

Step 5 of 5: Start KIVA!

Option A: Automatic (recommended) – Use the helper scripts:

Terminal 1

```
cd riverst  
./start_server.sh
```

Handles everything automatically

Terminal 2

```
cd riverst  
./start_client.sh
```

Checks server is running first

Option B: Manual – Run commands yourself:

Terminal 1

```
cd riverst/src/server  
conda activate riverst  
python main.py
```

Terminal 2

```
cd riverst/src/client/react  
npm run dev
```

Open Chrome: <http://localhost:5173>

Step 6: Select an Avatar and Go

- 1 On the homepage, click **“Select your avatar”**
- 2 Click on **Fabio** or **Bruce**
- 3 Go back to the homepage
- 4 Under **“Playground”**, click **“Demo ready to go”**
- 5 Click **“Run a demo”**
- 6 Allow microphone access when prompted
- 7 Start talking to your avatar!

- **“Failed to create session”** – Check OpenAI key in `src/server/.env`
- **Avatar not loading** – Clear browser cache or reset avatar in console
- **No microphone access** – Check Chrome permissions (Settings → Privacy → Microphone)
- **Port already in use** – Kill existing processes: `lsof -i :7860` or `lsof -i :5173`
- **ffmpeg not found** – `conda install -c conda-forge ffmpeg` or `brew install ffmpeg`
- **npm install fails** – Try `rm -rf node_modules && npm install`

Windows Users: GitHub Codespaces

KIVA doesn't run natively on Windows. Use GitHub Codespaces:

- 1 Go to <https://github.com/ebaenamar/KIVA>
- 2 Click green **Code** button → **Codespaces** tab
- 3 Click **Create codespace on main** (wait 2-3 min)
- 4 In terminal: `./start_server.sh` (Terminal 1)
- 5 New terminal: `./start_client.sh` (Terminal 2)
- 6 **Ports** tab → forward port 5173
- 7 Open the forwarded URL in browser

Free tier

60 hrs/month free

API Key

Set in `src/server/.env`

Quick Reference Card

 `git clone https://github.com/ebaenamar/KIVA.git`

 `cd KIVA/src/server && cp env.example .env` – Add OpenAI key

 `./start_server.sh` (Terminal 1)

 `./start_client.sh` (Terminal 2)

 Open `http://localhost:5173` in Chrome

That's it!

Questions?

 <https://github.com/sensein/riverst>

 Reach out on Slack if you get stuck